

Nicola Jackson

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EDUCATION

Boston University

Bachelor of Arts in Computer Science — GPA: 3.99

Expected May 2027

Boston, MA

- **Honors:** Dean's List (all semesters), CGS Merit Scholarship Recipient
- **Leadership & Activities:** Director of Engineering, Hack4Impact; Co-Founder & Vice Director, BU Quantum
- **Coursework:** Distributed Systems, Algorithms, Software Engineering, Databases, Data Structures, Computer Systems

EXPERIENCE

PNC Bank

Technology Intern

Jun. 2026 – Aug. 2026

Pittsburgh, PA

- Engineered Aria, an internal enterprise AI coding assistant designed to automate developer workflows and improve code delivery efficiency across enterprise engineering teams
- Promoted responsible AI practices by authoring 15+ prompt instruction and skill files for API scaffolding and database connectivity (MongoDB, Kafka, Oracle), improving code generation accuracy and adherence to security standards
- Reduced developer setup time by 75% by automating SDLC environment scaffolding and integrating structured validation workflows for AI-generated output

Hack4Impact BU

Director of Engineering (Dec. 2025 – Present) | Lead Software Engineer (May – Dec. 2025)

May 2025 – Present

Boston, MA

- Lead 3–5 cross-functional engineering teams (20+ developers) in delivering reliable full-stack web applications for non-profit clients, aligning tech solutions with social impact goals
- Drive end-to-end SDLC execution by defining system architectures, scoping roadmaps, and conducting rigorous code reviews to maintain code quality, security, and resiliency
- Collaborate with nonprofit stakeholders to gather requirements and translate them into scalable technical solutions, shipping 3-5 production applications per semester
- Designed and led SWE hiring pipeline, evaluating 50+ applicants through technical assessments and behavioral interviews
- As Lead SWE, led a 4-person team to rebuild the Hack4Impact website using React, Next.js, and Tailwind CSS

Boston University

Course Assistant, CS 111 (Intro to Computer Science)

Aug. 2025 – Present

Boston, MA

- Support 200+ students in learning Python through office hours, lab sessions, and detailed assignment feedback
- Debug student code and reinforce core concepts including control flow, data structures, and object-oriented programming

PROJECTS

BankBot | *Python, Streamlit, ChromaDB, Ollama*

Jul. 2025

- Developed a RAG-powered financial assistant using Python, LlamaIndex, and ChromaDB to retrieve official policy documents and provide accurate answers regarding fees, rates, and terms
- Integrated tool-calling with a local Llama 3.2 model to query live account balances and transaction histories while enforcing contextual prompt guardrails for accuracy
- Engineered keyword-triggered RAG routing and dynamic system-prompt injection (customer ID, selected account) to keep responses accurate and context-aware without unnecessary retrieval calls

Personal Portfolio | *React, JavaScript, MongoDB, Google Gemini API*

Mar. 2025

- Won **1st place** in a 60-person web design competition for UI/UX quality and interactivity
- Integrated a Google Gemini API-powered chatbot and real-time GitHub activity visualization to enhance user engagement
- Deployed a dynamic, responsive portfolio site showcasing projects and live development activity

Dawn2Dusk | *React, JavaScript, Google Gemini API, React Leaflet*

Mar. 2025

- Developed a full-stack application visualizing real-time global sunrise and sunset data on interactive maps
- Integrated natural language location queries using Google Gemini API and geospatial visualization with React Leaflet

Study Buddy | *JavaScript, Next.js, Tailwind CSS*

Aug. 2025

- Built a productivity web app with Pomodoro timer, task management, and deadline tracking features, taken solo from concept to deployment
- Optimized performance and responsiveness using Next.js server components and modern frontend practices

TECHNICAL SKILLS

Languages

Python, Java, SQL, C, Go, JavaScript, TypeScript

Frameworks & Libraries

React, Next.js, Node.js, Express.js, Tailwind CSS, NumPy

Databases & Tools

PostgreSQL, MongoDB, Git, GitHub, VS Code, Vercel

AI & Engineering Practices

AI-Assisted Development, Prompt Engineering, RAG Pipelines, Code Review, Agile, Ollama